4 Managing conversation and state in your application

This chapter covers

- Introducing AI SDK React server components
- Managing UI state in AI-powered applications
- Structured data generation using the Vercel AI SDK
- Tool and function calling with AI models

As we navigate the evolving landscape of AI-powered web applications, we encounter increasingly complex challenges that test the limits of reliability, performance, and scalability. What begins as a straightforward implementation often grows into a sophisticated system with complex interdependencies.

Consider, for example, how to perform state management in AI applications. The seemingly simple task of maintaining conversation history becomes a complex problem between client-side accessibility and server-side security. While storing messages on the client enables quick review, we can't implicitly trust client-provided data. This requires a dual-state approach, demanding meticulous synchronization between client and server to ensure data integrity and security.